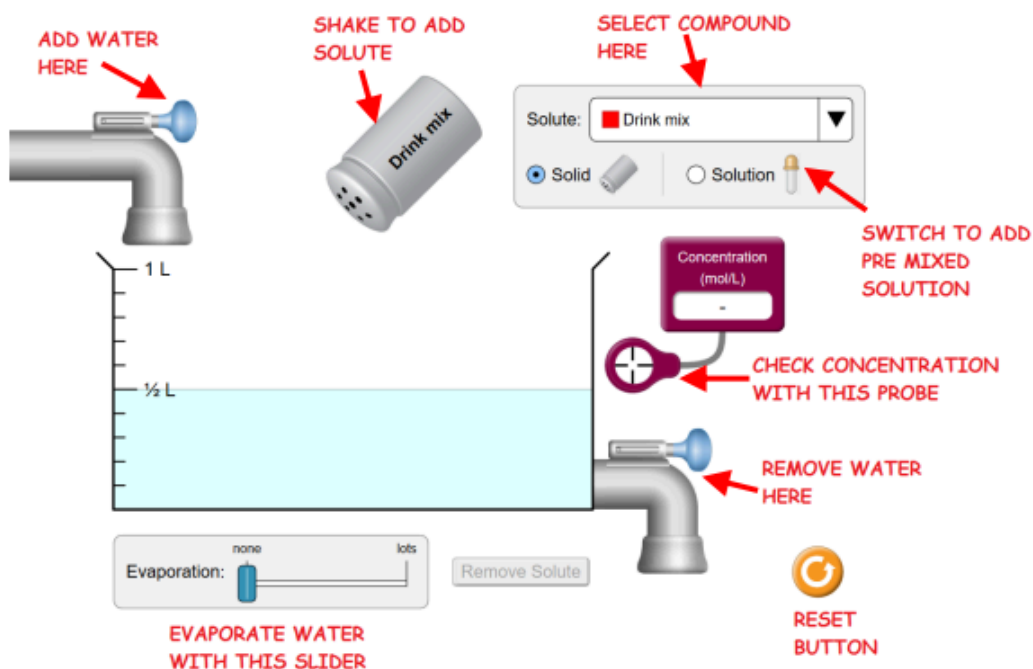


## University of Colorado Phet CONCENTRATION Exercise -

[https://phet.colorado.edu/sims/html/concentration/latest/concentration\\_en.html](https://phet.colorado.edu/sims/html/concentration/latest/concentration_en.html)



Familiarize yourself with the operation of the various functions of the simulation. You can always press the RESET button to get back to the original situation.

Press Reset. Move the Concentration Probe into the liquid. Then Shake some Drink mix into the water. Continue to add more solute.

### PART 1 – Effect of Changing Amount of Solute/Solvent

1. What happens to the concentration as more solute is added? got more red
2. What are the units for concentration on the measuring meter? mol/L
3. What happens to the color as more solute is added? becomes a more opaque red
4. What is the relationship between concentration and color? darker the color = higher concentration
5. Now add water. What happens to the concentration as the quantity of water increases? getting lower.

Explain why. (Mention what is happening to the quantity of solvent and solute)

Because the water (solvent) is increasing volume while the amount of solute stays the same. So the liquid becomes diluted.

6. Now drain some of the liquid using the valve on the lower right. What happens to the concentration as the quantity of water decreases? Stays the same

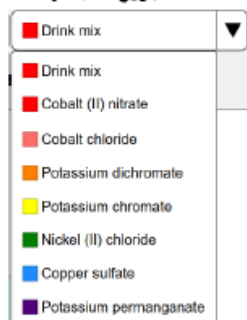
Explain why. (Mention what is happening to the quantity of solvent and solute)

It stays the same because equal amounts of the solvent & solute are being taken out since it's a mixed solution.

7. Now remove water with the Evaporation Slider Bar on the bottom. What happens to the concentration as the quantity of water decreases? increases

Explain why. (Mention what is happening to the quantity of solvent and solute)

Because with evaporation, only the water (solvent) goes down while the solute amount stays the same. So their ratio increases.



#### PART 2 - Saturation

Press the reset button. Switch to a different Compound by selecting the Drop Down Menu called SOLUTE. Select copper (II) sulfate. What is the formula for this compound?  $\text{CuSO}_4$

1. Can a saturated solution be created? yes

2. How do you know when a saturated solution is created?

When the solute stops mixing and just settles at the bottom.

3. Move the Concentration Probe into the liquid. (At this time the solution should be saturated) What is the concentration? (Include Units) 1.380 mol/L

4. Add more solute. What happens to the concentration? (Explain why)

Nothing happens because max capacity to dissolve a substance has already been reached.

#### PART 3 - Using the Eye dropper

Press the reset button. Switch to a the Eye dropper by selecting Solution in the upper right. Make sure you reselect copper(II) sulfate.



Logic Puzzle: Determine the concentration in the eye dropper. Write the steps below.

1. start with  $\frac{1}{2}$  L of water
2. add  $\text{CuSO}_4$  until solution is at 1 L
3. check concentration of solution.
4. multiply by 2 to get concentration of dropper

What is the concentration of the solution in the Eyedropper? 1 mol/L

Logic Puzzle: How can a saturated solution be made from the liquid from the eye dropper? Write the steps below.

You can do this using the evaporation slider so the amount of solvent goes down but solute stays the same.

#### PART 4 - Controlling the Concentration

Press the reset button. Add enough copper(II) sulfate to create a 0.5 Molar Solution in the  $\frac{1}{2}$  liter of water originally in the beaker.

1. Show the calculation of the number of moles of in the beaker. (You may need the Equation for Molarity, look it up.)

$$0.5 \text{ mol/L} \cdot 0.5 \text{ L} = 0.25 \text{ mol}$$

2. **Predict** the concentration if the solution is diluted to a volume of 1 liter. Show the calculation of this concentration, include units in all your work.

$$M = \frac{0.25 \text{ mol}}{1.00 \text{ L}} = 0.25 \text{ mol/L}$$

After you make the calculation: Add water with the valve in the upper left until there is 1 liter of water in the beaker.

3. What is the concentration? 0.25 mol/L Does this match your calculation? Yes or No

4. **Predict** the concentration if the solution is now concentrated (by using evaporation) to a volume of .75 liters (1/2 way between .5 liters and 1 liter). Show the calculation of this concentration, include units in all your work.

$$M = \frac{0.25 \text{ mol}}{0.75 \text{ L}} = 0.333... \text{ mol/L}$$

**After you make the calculation:** Evaporate water with Evaporation Slider Bar at the bottom until there is .75 liter of solution in the beaker.

5. What is the concentration? 0.333 mol/L Does this match your calculation? Yes or No

6. From the concentration & Volume **calculate** the number of moles of solute in the solution. Show your calculation with units.

$$0.333 \text{ mol/L} \cdot 0.75 \text{ L} = 0.25 \text{ mol}$$

7. How does the # moles of solute compare to the # of moles of solute calculated in question 1. **Explain why these numbers compare as they do.**

It is the same because we haven't actually changed the amount of solute this whole time.

8. **Predict** the concentration & # of moles of solute if water is added until the volume is 1 liter. (This should be easy)

$$M = \frac{0.25 \text{ mol}}{1.00 \text{ L}} = 0.25 \text{ mol/L}$$

9. **Fill the beaker** to a volume of 1 liter and check the concentration. What is the concentration? 0.250 mol/L

Open the Remove Water Valve (on the lower right) until there is  $\frac{1}{2}$  a liter of solution in the beaker.

10. What happened to the concentration when  $\frac{1}{2}$  of the solution was allowed to run out? (Explain why in terms of the amount of solute and solvent)

Stayed the same because the solvent and solute left in the same ratio, so the concentration didn't actually change

11. From the concentration & volume calculate the number of moles of solute in the solution. Show your calculation with units.

$$0.25 \text{ mol/L} \cdot 0.50 \text{ L} = 0.125 \text{ mol}$$

12. How does the # moles of solute compare to the # of moles of solute calculated in question 1. Explain why these numbers compare as they do.

It's half of the # moles of solute because half of the solution was drained out (aka half of solute was taken out).

13. Predict the concentration if the solution is diluted to a volume of 1 liter. Show the calculation of this concentration, include units in all your work.

$$M = \frac{0.125 \text{ mol}}{1.000 \text{ L}} = 0.125 \text{ mol/L}$$

After you make the calculation: Add water with the valve in the upper left until there is 1 liter of water in the beaker.

14. What is the concentration? 0.125 mol/L Does this match your calculation? Yes or No

#### SUMMARY OF LEARNING OBJECTIVES

Adding solute (solid) to an unsaturated solution causes the concentration of the solution to:

INCREASE / DECREASE / REMAIN UNCHANGED.

Adding pure water to a saturated solution will cause the concentration of the solution to:

INCREASE / DECREASE / REMAIN UNCHANGED.

Adding a solid salt to a saturated solution causes the concentration of the solution to:

INCREASE / DECREASE / REMAIN UNCHANGED.

Evaporation acting on an unsaturated solution causes the concentration of the solution to:

INCREASE / DECREASE / REMAIN UNCHANGED

Evaporation acting on a saturated solution causes the concentration of the solution to:

INCREASE / DECREASE / REMAIN UNCHANGED.